

SFT V11.1: The Planck-Area Accounting Lemma

Why the Planck Length Matters, and Why It Is Not a Lattice

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April 2026

Abstract

Every quantum gravity program assigns meaning to the Planck length. Loop quantum gravity says spacetime is a lattice at that scale. String theory says it vibrates. Causal set theory says it is sprinkled with discrete events. Each program introduces new structure at 10^{-35} meters to explain why that scale matters.

Solvency Field Theory does not introduce new structure. In SFT, the Planck area $\ell_P^2 = \hbar G/c^3$ is the elementary unit of holographic boundary capacity — the geometric cost of one bit. A particle ticking at its Compton frequency writes one bit per cycle, and each bit demands one Planck area of boundary. The Planck length is not a lattice spacing. It is the size of one ledger entry. Local concentrations of ledger entries produce gravity. The global sum of ledger entries produces expansion. This note derives these statements, shows that the same accounting recovers both the Newtonian potential and the SFT Friedmann ratio $H_{\text{SFT}}^2/H_{\text{std}}^2 = 1/(4\pi^2)$, and explains why no invisible lattice is needed.

1 The Question

The Planck length is

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35} \text{ m.} \quad (1)$$

It is built from three constants: $\hbar$ (quantum mechanics), G (gravity), and c (relativity). It is the only length you can construct from these three constants, and it is unimaginably small — roughly 10^{20} times smaller than a proton. Every approach to quantum gravity agrees that something important happens at this scale, but they disagree about what.

The question this note answers is: what does the Planck length mean in Solvency Field Theory?

The answer is not “spacetime is discrete at this scale.” Spacetime curvature is a coarse-grained field and can be made arbitrarily small by going far enough from any source. There is no minimum curvature. The answer is:

The Planck area is the elementary boundary-accounting increment. A single maintenance tick — one complete Compton phase cycle — demands one Planck-area unit of holographic boundary capacity. Curvature is the coarse-grained geometric response to many such increments accumulated locally. Expansion is the coarse-grained geometric response to all such increments summed globally.

The Planck length is the pixel size of the ledger. Not the pixel size of space.

2 One Tick, One Bit, One Planck Area

A particle with energy E is serviced at the rate

$$\dot{N} = \frac{E}{h_P}, \quad (2)$$

where h_P is Planck's constant. For a particle at rest with mass m :

$$\dot{N}_C = \frac{mc^2}{h_P}. \quad (3)$$

Each tick writes one bit to the holographic boundary. Each bit occupies one elementary capacity unit. The capacity of the boundary is measured in Planck areas:

$$I = \frac{A}{4\ell_P^2}, \quad (4)$$

where A is the boundary area. So the elementary capacity increment associated with one tick is

$$\delta A = \xi \ell_P^2, \quad (5)$$

where ξ is a convention factor that tracks the bit/nat normalization. The mechanism does not depend on the choice of ξ ; the convention controls the bookkeeping units, not the physics.

The boundary-area demand rate of a single particle is then

$$\dot{A}_{\text{tick}} = \xi \ell_P^2 \cdot \frac{mc^2}{h_P}. \quad (6)$$

Substituting $\ell_P^2 = \hbar G/c^3$ and $h_P = 2\pi\hbar$:

$$\dot{A}_{\text{tick}} = \xi \left(\frac{\hbar G}{c^3} \right) \left(\frac{mc^2}{2\pi\hbar} \right) \quad (7)$$

The $\hbar$ cancels. What remains is

$$\boxed{\dot{A}_{\text{tick}} = \frac{\xi G m}{2\pi c}} \quad (8)$$

This is the Planck-area accounting lemma in one line. A particle's mass m sets its Compton tick rate. Its tick rate sets its boundary-area demand. G converts the maintenance cost into geometry. c sets the causal processing scale. And 2π appears because SFT counts complete cycles, not radians — the same phase factor that appears in the $4\pi^2$ spine.

Every ingredient of the theory is in this one equation. Mass is ticking rate. Ticking rate is boundary demand. Boundary demand is geometric cost. Geometric cost is measured in Planck areas.

3 What Is Quantized, and What Is Not

This is the most important distinction in the note, and it must be stated precisely.

What is quantized is the **accounting**. One tick, one capacity increment, $\delta A = \xi \ell_P^2$. You cannot write half a bit. You cannot demand half a Planck area. The ledger is discrete.

What is **not** quantized is the **curvature**. The curvature at any point in space is the coarse-grained response to an enormous number of accounting increments from many sources accumulated over time. At macroscopic distances, the curvature produced by a single particle is

$$R(r) \sim \frac{Gm}{c^2 r^3}, \quad (9)$$

which can be made arbitrarily small by increasing r . There is no minimum curvature value.

The analogy is a digital photograph. Each pixel is discrete — it stores a finite number of color values. But the photograph itself depicts smooth gradients, subtle shadows, continuous curves. The smoothness is not an illusion; it is the macroscopic appearance of a very large number of discrete elements. The individual pixel is not the minimum visual feature. It is the minimum accounting unit of the representation.

Likewise, the Planck area is not the minimum curvature of spacetime. It is the minimum accounting unit of the holographic boundary. Smooth geometry is what the accounting looks like at scales much larger than the pixel.

4 Many Ticks Locally: Gravity

Now fill a region of space with many particles, each ticking at its own Compton rate. The total maintenance load per unit volume is the operation density:

$$\rho_{\text{ops}} = \frac{\varepsilon + 3p}{h_P}, \quad (10)$$

where ε is energy density and p is pressure. This is the continuum limit of the discrete particle sum, derived in V10.10 by counting individual tickers and their transport burden.

In the weak-field limit, with pressureless matter ($p = 0$), define the dimensionless lag potential $\psi = \Phi_N/c^2$, where Φ_N is the Newtonian potential. The accounting map from operation density to lag curvature is the Poisson equation:

$$\boxed{\nabla^2 \psi = \frac{4\pi G h_P}{c^4} \rho_{\text{ops}}} \quad (11)$$

This says: the local concentration of maintenance ticks sources the local curvature of the lag field. More ticks per unit volume means steeper curvature. The operation density ρ_{ops} is the tick concentration. The Poisson equation is the accounting rule that converts tick concentration into geometric slope.

For a single particle at the origin with $\varepsilon(\mathbf{x}) = mc^2 \delta^3(\mathbf{x})$, the Green's function solution is

$$\psi(r) = -\frac{Gm}{c^2 r}, \quad (12)$$

and the gravitational acceleration is

$$g = -\nabla \Phi_N = -\frac{Gm}{r^2} \hat{r}. \quad (13)$$

Newton's law. Not postulated — recovered from the accounting. A local concentration of maintenance ticks creates a lag gradient. The gradient is gravity.

5 All Ticks Globally: Expansion

Now instead of looking at the local gradient, sum all the ticks in the entire observable universe.

The Hubble volume is

$$V_H = \frac{4\pi c^3}{3H^3}. \quad (14)$$

The total operation rate inside the Hubble sphere, for homogeneous energy density ε , is

$$\dot{N}_{\text{tot}} = \frac{\varepsilon V_H}{h_P}. \quad (15)$$

Over one Hubble time $t_H = 1/H$, the total processed count is

$$N_H = \frac{\varepsilon V_H}{h_P H}. \quad (16)$$

The holographic capacity of the Hubble boundary is

$$I_H = \frac{A_H}{4\ell_P^2} = \frac{\pi c^2}{\ell_P^2 H^2}. \quad (17)$$

The settlement condition — the ledger balances over one Hubble time — is

$$N_H = I_H. \quad (18)$$

Solving for H^2 and substituting $\ell_P^2 = \hbar G/c^3$ and $h_P = 2\pi\hbar$ gives

$$H_{\text{SFT}}^2 = \frac{2G\varepsilon}{3\pi c^2}. \quad (19)$$

Dividing by the standard Friedmann equation $H_{\text{std}}^2 = 8\pi G\varepsilon/(3c^2)$:

$$\boxed{\frac{H_{\text{SFT}}^2}{H_{\text{std}}^2} = \frac{1}{4\pi^2}} \quad (20)$$

The same accounting that produces gravity locally produces the SFT Friedmann ratio globally. The Planck area appears in both: locally as the granularity of the lag field's accounting, globally as the capacity unit of the holographic boundary. The $4\pi^2$ appears because both the local and global conversions pass through the same two factors of 2π — one from complete-cycle counting, one from horizon capacity.

6 The Same Mechanism in One Diagram

The entire content of this note reduces to three lines:

$$\text{One tick} \implies \delta A = \xi \ell_P^2$$

$$\text{Many ticks locally} \implies \nabla^2 \psi = \frac{4\pi G \hbar_P}{c^4} \rho_{\text{ops}} \implies \mathbf{gravity}$$

$$\text{All ticks globally} \implies N_H = I_H \implies \frac{H_{\text{SFT}}^2}{H_{\text{std}}^2} = \frac{1}{4\pi^2} \implies \mathbf{expansion}$$

Gravity is what the accounting looks like when ticks concentrate. Expansion is what the accounting looks like when ticks are summed. The Planck area is the unit of accounting in both cases. One mechanism, two scales, one conversion factor.

7 No Invisible Lattice

This is why SFT does not need a quantum gravity program.

Loop quantum gravity introduces a discrete spatial lattice at the Planck scale — spin networks and spin foams that replace smooth spacetime with a combinatorial structure. String theory introduces one-dimensional strings whose oscillation modes produce particles and gravity. Causal set theory introduces a random sprinkling of discrete events. Each program adds microscopic furniture to explain what happens at 10^{-35} meters.

SFT adds nothing. The Planck area is not a lattice cell embedded in space. It is the accounting scale of the holographic boundary. The smooth metric is not a continuous substance that must be discretized by adding new structure. The smooth metric is the macroscopic geometry of an enormous accounting process — what 10^{122} ledger entries look like when you zoom out.

The distinction can be stated precisely:

$$\ell_P^2 \text{ is a \textbf{boundary capacity unit}, not a \textbf{bulk lattice cell}.} \quad (21)$$

A lattice is extra structure. A ledger entry is a process. The Planck length is the scale of the process, not the scale of a thing. You do not need to quantize spacetime if spacetime was never a continuous substance in the first place — if it was always the accumulated geometric cost of discrete information processing, one tick at a time, one Planck area at a time, from the beginning.

8 What This Note Establishes

The Planck-area accounting lemma establishes the following chain:

1. A particle ticks at mc^2/h_P . That is existence.
2. Each tick demands one Planck-area unit of boundary capacity. The demand rate is $\dot{A}_{\text{tick}} = \xi Gm/(2\pi c)$.
3. What is quantized is the accounting increment, not the curvature. Curvature is the coarse-grained response to many increments and can be arbitrarily small.
4. Local concentrations of ticks source the weak-field lag potential through the Poisson equation. The gradient of the lag potential is gravity.
5. The global sum of all ticks, settled against the Hubble boundary capacity, produces the SFT Friedmann ratio $H_{\text{SFT}}^2/H_{\text{std}}^2 = 1/(4\pi^2)$.
6. No invisible lattice is required. The Planck area is the boundary accounting unit, not a bulk spatial cell. Smooth geometry is what an enormous ledger looks like from far away.

The Planck length matters in SFT not because space is made of Planck-sized blocks, but because information is discrete and geometry is the cost of processing it. One tick, one bit, one Planck area. That is the pixel. Everything else is the photograph.

Portions of the mathematical derivation and manuscript preparation were developed in collaboration with Claude/Opus (Anthropic), ChatGPT/5.4 and 5.5 (OpenAI), Gemini/Ender (Google), and Grok (xAI).